DELTA CLIMATE RESEARCH

Project Build Blueprint: Institutional Innovation & Frontier Design

1. CORE IDENTITY & AESTHETIC

Delta Climate Research is a boutique consultancy at the intersection of climate intelligence, systems thinking, and frontier design. The website must feel like a **high-fidelity dashboard for planetary survival**.

- **Visual Reference:** Dark Matter Labs (Institutional Density) + ClimateCulture (Narrative Boldness) + Field.blue (Generative Art).
- **Color Palette:** Base: #080808 (Deep Onyx). Accent: #00F2FF (Bioluminescent Cyan) & Metallic Bronze.
- **Typography:** Pair a technical sans (Inter / SF Pro) with a sharp editorial serif (Editorial New) for headers.

2. TECHNICAL STACK & ARCHITECTURE

Prioritize speed, SEO, and smooth rendering of generative graphics.

- **Framework:** Next.js (App Router) or Astro (for maximum static performance).
- **Motion/FX:** Three.js (Background Shader), Framer Motion (UI transitions).
- **Styling:** Tailwind CSS (Utility-first, pixel-perfect borders).

```
// Suggested Global Grid System (Tailwind)
<div class="grid grid-cols-1 md:grid-cols-12 min-h-screen border-t border-white/10">
    <aside class="col-span-3 border-r border-white/10 p-8 sticky top-0 h-screen">
        <!-- Filter Navigation -->
    </aside>
    <main class="col-span-9 divide-y divide-white/10">
        <!-- Systems Feed -->
    </main>
</div>
```

3. THE SYSTEMS FEED (UI COMPONENT SPECS)

This is the engine of the site. It must display dense research with extreme clarity.

Element	Description/Behavior
Outer Wrapper	1px solid border-bottom. Padding: 40px top/bottom. Hover bg: rgba(255,255,255,0.02).
Meta-Data Header	ID (e.g., [01]), Date, and Category in Monospace 8pt Cyan text.
Typography	Title: Serif, 24pt. Excerpt: Sans, 10pt, Color: #909090. Max-width: 60ch.

INTERACTION STATES

- **Focus Pull:** On hover, blur all siblings by 1.5px and drop their opacity to 0.5.
- **The Kinetic Arrow:** The [↗] icon should translate (x:3px, y:-3px) on hover using a spring transition.

4. GENERATIVE BACKGROUND (FIELD.BLUE INSPIRED)

The "Fluid Background" is an interactive WebGL shader representing climate currents.

SHADER IMPLEMENTATION STRATEGY:

Use a **Simplex Noise** algorithm to deforming a vertex grid. The "fluid" effect comes from time-based offsets and cursor position inputs.

```
// Vertex Shader snippet logic
uniform float uTime;
uniform vec2 uMouse;
varying float vElevation;

void main() {
    vec4 modelPosition = modelMatrix * vec4(position, 1.0);
    float elevation = sin(modelPosition.x * 3.0 + uTime) * sin(modelPosition.z * 2.0 +
uTime) * 0.15;
    modelPosition.y += elevation;
    // Add Mouse distortion
```

```
    float dist = distance(modelPosition.xz, uMouse);
    modelPosition.y += (1.0 / (dist + 0.5)) * 0.1;

    gl_Position = projectionMatrix * viewMatrix * modelPosition;
}
```

5. CONTENT STRATEGY & PLACEHOLDERS

- **Intelligence:** Innovation White Papers (Nature-based solutions, Blue Carbon).
- **Partnerships:** Strategic Sectoral collaborations (Industrial/Global).
- **Creative Studio:** Speculative Design, Geospatial Twin concepts, Biophilic architecture.